

Positive-Velocity Goldfish Flow: Global Root Interlacing and One-Carrier Scattering

Route-A source-local certificate HCS-C315

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Abstract

For arbitrary ordered real positions and strictly positive velocities, we prove that the Calogero goldfish polynomial pencil has simple real roots at every real time. A decreasing partial-fraction map gives exact signed-time interlacing, global collision avoidance, and strict coordinate monotonicity. The fixed roots of the invariant polynomial become scattering anchors: we derive their positive first inverse-time coefficients and, by an exact Vieta law, the intercept and first correction of the unique ballistic root. The carrier moves from the leftmost incoming rank to the rightmost outgoing rank, with sharp failures on the boundary of the positive cone. Exact rational, Sturm, high-precision, replay, and hostile-mutation evidence audits the formulas while preserving classical ownership and the strict Route-A boundary.

1 Polynomial linearization and global interlacing

Fix $N \geq 2$, positions $x_1 < \dots < x_N$, and velocities $v_i > 0$. Define

$$P(z) = \prod_{i=1}^N (z - x_i), \quad Q(z) = \sum_{i=1}^N v_i \prod_{j \neq i} (z - x_j), \quad p(z, t) = P(z) - tQ(z). \quad (1)$$

Let $y_1 < \dots < y_{N-1}$ denote the roots of Q .

Theorem 1 (Complete positive-cone flow). *For every $t \in \mathbb{R}$, $p(\cdot, t)$ has N simple real roots $z_1(t) < \dots < z_N(t)$. They obey*

$$t > 0: \quad x_i < z_i(t) < y_i \quad (i < N), \quad x_N < z_N(t), \quad (2)$$

$$t < 0: \quad z_1(t) < x_1, \quad y_i < z_{i+1}(t) < x_{i+1} \quad (i < N). \quad (3)$$

Every $\dot{z}_i$ is positive, $\sum_i \dot{z}_i = V := \sum_i v_i$, and

$$\ddot{z}_i = 2 \sum_{j \neq i} \frac{\dot{z}_i \dot{z}_j}{z_i - z_j}. \quad (4)$$

Thus (1) is a complete collision-free goldfish solution with $z_i(0) = x_i$ and $\dot{z}_i(0) = v_i$.

Proof. Off the poles x_i , partial fractions give

$$R(z) := \frac{Q(z)}{P(z)} = \sum_{i=1}^N \frac{v_i}{z - x_i}, \quad R'(z) = - \sum_{i=1}^N \frac{v_i}{(z - x_i)^2} < 0. \quad (5)$$

Hence $R = 0$ has exactly one root y_i in each interval (x_i, x_{i+1}) . When $t \neq 0$, $p(z, t) = 0$ is equivalent to $R(z) = 1/t$. Strict decrease and the one-sided pole limits give exactly the intervals in (2)–(3), including the sign-selected exterior interval. All roots are simple. At $t = 0$ they are the simple roots of P , so this construction covers all real time.

For $t \neq 0$, differentiation of $R(z_i(t)) = 1/t$ gives

$$\dot{z}_i = - \frac{1}{t^2 R'(z_i)} > 0. \quad (6)$$

At zero, implicit differentiation instead gives $\dot{z}_i(0) = Q(x_i)/P'(x_i) = v_i$. The coefficient of z^{N-1} in p gives the exact law

$$\sum_i z_i(t) = \sum_i x_i + Vt, \quad (7)$$

and therefore $\sum_i \dot{z}_i = V$. Differentiating $p(z, t) = \prod_i (z - z_i(t))$ in t shows that the time-independent Q is

$$Q(z) = \sum_i \dot{z}_i(t) \prod_{j \neq i} (z - z_j(t)). \quad (8)$$

At a root z_i , elementary logarithmic differentiation gives

$$\frac{p_{zz}}{p_z} = 2 \sum_{j \neq i} \frac{1}{z_i - z_j}, \quad (9)$$

$$\frac{Q'}{p_z} = \sum_{j \neq i} \frac{\dot{z}_i + \dot{z}_j}{z_i - z_j}. \quad (10)$$

Twice differentiating $p(z_i(t), t) = 0$ and inserting (9)–(10) yields (4). Finally, $p(z, t + s) = p(z, t) - sQ(z)$ proves the exact flow group law. $\square$

2 One-carrier scattering and boundary sharpness

Put

$$M = \sum_i v_i x_i, \quad c = \frac{M}{V}, \quad \beta_i = -\frac{P(y_i)}{Q'(y_i)}, \quad B = \sum_{i=1}^{N-1} \beta_i. \quad (11)$$

Theorem 2 (Two-ended rank-transfer scattering). *Every β_i is positive. As $t \rightarrow -\infty$,*

$$z_1(t) = Vt + c + \frac{B}{t} + O(t^{-2}), \quad (12)$$

$$z_{i+1}(t) = y_i - \frac{\beta_i}{t} + O(t^{-2}) \quad (i < N), \quad (13)$$

whereas, as $t \rightarrow +\infty$,

$$z_i(t) = y_i - \frac{\beta_i}{t} + O(t^{-2}) \quad (i < N), \quad (14)$$

$$z_N(t) = Vt + c + \frac{B}{t} + O(t^{-2}). \quad (15)$$

All expansions differentiate termwise. Thus the incoming velocity vector tends to $(V, 0, \dots, 0)$ and the outgoing vector to $(0, \dots, 0, V)$ in rank order.

Proof. At a simple zero y_i of Q , the implicit-function theorem with $s = 1/t$ applied to $sP(z) - Q(z) = 0$ gives

$$z = y_i + \frac{P(y_i)}{tQ'(y_i)} + O(t^{-2}) = y_i - \frac{\beta_i}{t} + O(t^{-2}). \quad (16)$$

Since $y_i \in (x_i, x_{i+1})$, the signs of $P(y_i)$ and $Q'(y_i)$ are opposite; hence $\beta_i > 0$. Comparing the two leading coefficients in $Q = V \prod_i (z - y_i)$ yields

$$\sum_{i=1}^{N-1} y_i = \sum_{i=1}^N x_i - c. \quad (17)$$

Subtracting all finite expansions (16) from the exact sum (7), and using (17), gives the ballistic expansions including the B/t term. Interlacing (2)–(3) fixes their ordered ranks. Analyticity in s justifies differentiated remainders and completes the velocity claim. $\square$

The strict positive cone is sharp. For $x = (0, 1)$ and $v = (1, 0)$, $p = (z - 1)(z - t)$ collides at $t = 1$. For $v = (1, -1)$, $p = z^2 - z + t$ has a double root at $t = 1/4$ and nonreal roots afterwards. The $N = 1$ face is a free particle; coincident initial positions are singular; all-negative velocities follow by time reversal but are not merged into Theorems 1–2.

3 Evidence, collisions, and Route-A boundary

Fourteen rational cases in dimensions two through seven supply 98 signed-time rows, 413 root-time cells, 45 fixed anchors, 28 asymptotic rows, and 1,892 audited evidence leaves. An independent checker performs 2,047 checks; SymPy closes seven identity groups; isolated replay is byte exact; and 45 repaired-hash, stale-hash, strict-parser and optimized-Python attacks must fail. Exact Sturm intervals certify every sampled real-root statement. The evidence regresses formulas; the proofs own arbitrary finite N and untruncated time.

C196 treats repulsive inverse-square Calogero–Moser motion through a Hermitian eigenvalue pencil and N asymptotic free velocities. C292 uses sticky merger resolution, while C296 uses event-driven elastic hard-rod relabeling. None owns this smooth velocity-coupled pencil or its one-carrier positive-cone scattering theorem.

Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the strict tuple is

(`A0_FAIL`, `A1_FAIL`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`).

The arbitrary continuous data have no prime owner or logarithmic clock; strict monotonicity supplies no primitive periodic ledger; and no target determinant or Weil compression is defined. Polynomial linearization is only a formal A4 hint, not a canonical self-adjoint quantization. Route A is rejected and Route B remains locked. No target local data, Euler factors, root number, automorphy, divisor/counting law, functional equation, zero match, or Hilbert–Pólya operator is asserted.

AI use. A generative language model assisted drafting and code scaffolding. The displayed proofs, independent reconstruction, hostile tests and deterministic artifacts define the audit record.

Source lineage

References

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